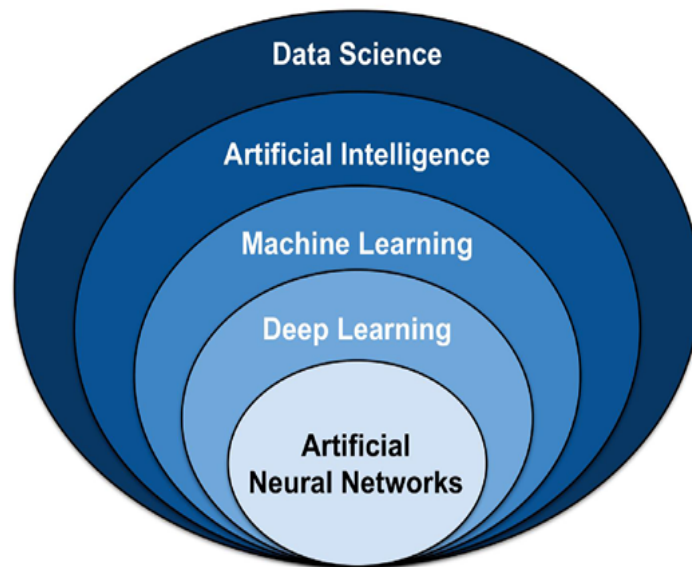


The Origins of Artificial Intelligence:

Over the past decade AI has become a popular field both within and outside scientific community. In 1956 computer scientists proposed an idea that computers can be programmed to think and reason, “that every aspect of learning or any other feature of intelligence, in principle, be so precisely described that a machine could be made to simulate it”. They described this principle as “artificial intelligence.”

AI encompasses techniques other than ‘learning’; for instance, there is the so called *‘symbolic’ approach to AI* in which particular functions are programmed to perform certain tasks using human knowledge. For instance, individuals can set special limits of heating or cooling for the room temperature regulating. This is due to the fact that while logical and well defined problems are modelled by difficulties of symbolic AI, for more complex problems such as speaking or facial recognition the algorithms of machine learning (ML) or deep learning DL are used.



Why Need Of Machine Learning:

Artificial Intelligence (AI) consists of methods for making computers mimic human functions and performance and make decisions on their own with aspects such as reasoning, learning, planning, perceiving and communicating [1](Russell and Norvig 2021). The initial AI work applied the use of rules and knowledge bases methods but these fell short of expectations because humans could not clearly express implicit knowledge [2](Chen et al. 2008). The pros of the Machine Learning (ML) include, it applies algorithms that learn from data and perform better with experience, and not programming like traditional techniques. ML is very useful in solving exercises that deal with high dimensions such as classifying, regression as well as clustering and has been adopted in fraud detection, credit rating, speech and image recognition together with natural language processing [3](Bishop 2006).

Reference (Machine learning and deep learning from Springer Link) – 8 April 2021

Based on the given problem and the available data, we can distinguish three types of ML:

1) Supervised Learning:

In supervised learning the input data is given with the correct labels and the model learns and train from that data then uses this data to make predictions about new and unseen data. For eg if a child is given some fruits(data) and the names(label) of fruits are also taught to the child. Eventually child will start to recognize fruits without telling the names of the fruit.

2) Unsupervised Learning:

In unsupervised learning only the data is given without the labels. The model sorts and groups the inputs based on similarities like size, color, or shape, so it finds patterns on its own without any prior knowledge.

3) Reinforced Learning:

In reinforced learning agent learns by taking actions and interacting with the environment. And based on the results it continuously improves itself to maximize the possibility to get the most optimized result.

Reference (Master Supervised And Unsupervised Learning Algorithms With real Examples – First Edition) – 2022

Introduction to Image Recognition:

It is estimated 75% of the information received by a human is visual. When you receive and use this visual information, we refer this process as sight, perception or understanding. When a computer receives and uses this visual information we call this computer image processing or recognition.

Computers traditionally had difficulties with understanding these visual information or images. Thanks to the new image recognition technology, we now have specific software and applications that can interpret visual information. The platform is based on a number of deep learning functionalities with especial emphasis to convolutional neural networks (CNNs) in order to process visual data. These models excel at extracting, classifying, and analyzing features within images, achieving remarkable accuracy.

Difference between Image Recognition vs. Image Detection:

Image Detection means to detect an object without knowing about the object for eg (object name, and its description). It will only identify the object for eg if the image of dog is shown it will detect dog as an object without knowing it that it's a dog whereas in Image recognition it recognizes the image for eg if the images of animals are shown it will not only detect the animals but also recognize if the animal is lion or a tiger etc.

Reference (AI Image Recognition in 2024 – New Examples and Use Cases from Addepto) - July 24, 2024

What is Image Recognition Used for?

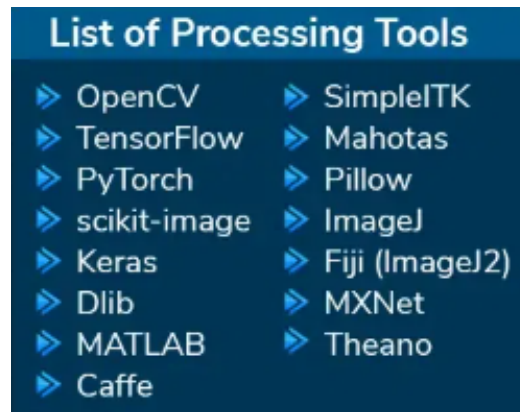
In all industries, AI image recognition technology applications provide economic value in industries such as healthcare, retail, security, agriculture, and many more. Some of its application that are used in daily life are mentioned below:

- Face Analysis
- Pattern and Object Detection
- Autonomous Vehicles
- Medical Image Analysis
- Security and Surveillance
- Crop Monitoring:

Image Processing Tools:

A key Machine learning component is Image Processing, which allows extracting useful information from images for tasks like segmentation, object detection, and image classification.

Top image-processing tools for Machine Learning exploring their unique features, capabilities, and the roles they play in enhancing machine-learning models.



Role of ML and DL in Image Recognition:

In recent years, deep learning technology has been used in various sectors. Deep learning has developed human-like abilities, such as knowledge learning, problem-solving, and decision-making. [1] Big companies have tried to adopt the latest digital technologies.

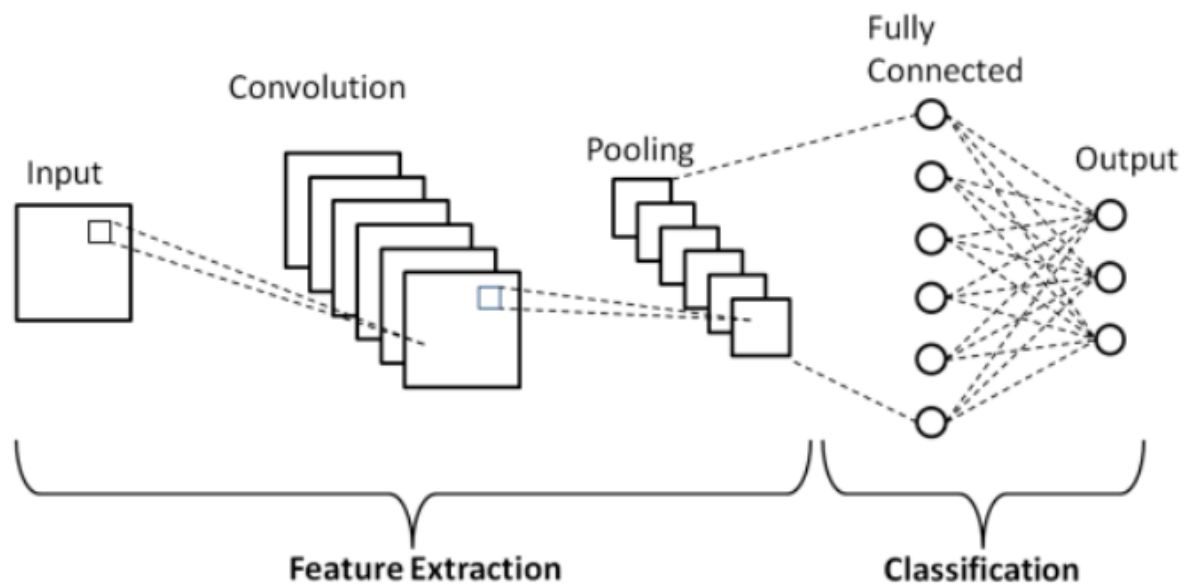
In general, machine learning and deep learning can perform self-training without repetitive programming by humans. Deep learning requires initial data collection, called a data set, to predict the outcome of the data. Deep learning will produce output data based on training and testing data. After passing the learning evaluation, deep learning can predict data [4].

One of the most widely used deep learning algorithms is the convolutional neural network (CNN). CNN was first introduced in the 1960s [10] and has shown promising performance results in computer vision. CNN has become the most representative neural network in deep learning [12]. CNN has been utilized to solve complicated visual tasks with high computation [11] and is mainly used in image classification [13], [14], segmentation, object detection and video processing.

Convolutional Neural Network (CNN):

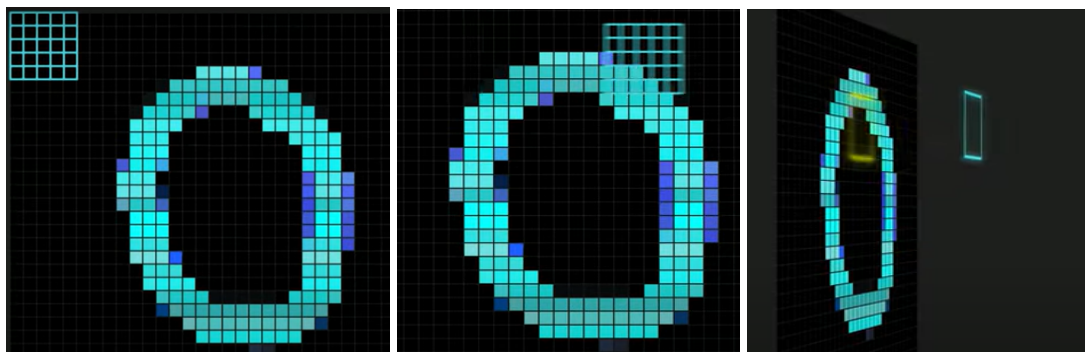
CNN has four layers:

i) Convolution layer ii) pooling layer iii) Fully connected layer iv) nonlinearity layer.



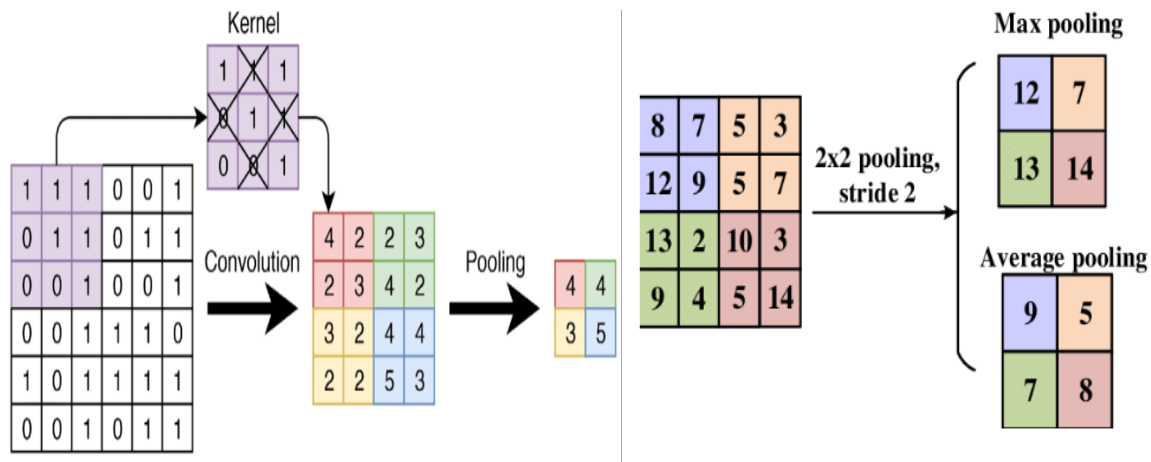
1) Convolutional Layer:

A machine understands an image as an arrangement of numbers which is also referred to as a matrix. Every number denotes the light intensity at a specific location called a pixel. The convolutional layers are the first layers which have 1 or multiple filters/kernels which are used to specify features in an image, such as edges, textures or patterns. The filter is a small grid (for instance, it can be 3x3 or 5x5) of numbers that is applied to the image through a particular process called convolution. The convolutional layer captures details from the image.



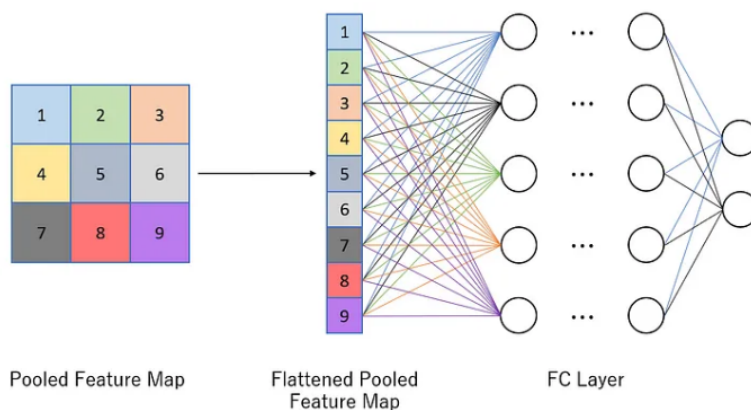
2) Pooling Layer:

It takes all the inputs from convolutional layer and pools them and reduces them. The pooling layers are an efficient way of reducing the size of the feature maps. The layer summarises the features present in a region of the feature map generated by a convolution layer. Thus, the subsequent operations are performed on the summarized features, not on the features generated at specific locations by the convolution layer. There are 2 types of pooling max and avg pooling.



3) Fully Connected Layer:

The third layer is the last layer which is the fully connected layer, commonly called the convolutional output layer/dense layer. These layers are termed “fully connected” since each neuron in one layer has feed forward connections to every neuron in the next layer forming a highly interconnected network.. It receives input from the final pooling or the convolutional output layer, before being flattened and fed into the next layer. They are responsible for producing the final output predictions of a neural network.



4) Nonlinearity Layer (Activation Function):

Nonlinearity Layer (Activation Function): An activation function is an important part of CNN layers. The filtered output provides another mathematical function called an activation function [26]. ReLU, which is the abbreviated form of Rectified Linear Unit [27], is the most commonly used activation function in feature extraction using CNN. The sole aim of the activation function is to determine the output of a neural network, for instance, yes or no. The activation function converts the output values to the range of -1 and 1, 1 and 0 and so on.

Reference(Understanding of Convolutional Neural Network (CNN) From ASCEE.ORG) – Jan 15 2023

Challenges in Image Recognition:

The field of image recognition has witnessed tremendous progress in recent years, fueled by groundbreaking architectures AlexNet [1], ResNet [2], ViT [3] and Swin family [4]. These architectures excel at processing datasets like ImageNet [5], which primarily contain natural images with resolutions well below one megapixel. However, many critical applications rely on the analysis of high-resolution images to extract details about objects of interest. For eg Self-driving cars to identify traffic signs far away OR Imaging to distinguish minute structures of tissues or cells in order to determine a diagnosis. However, working with high-resolution images (very detailed, large images) brings major challenges:

1) High Memory and Computational Needs:

- The memory and processing power are much large for larger images as size of image increases.
- This usually can't be handled by small devices such as the laptop or small GPU'S.

2) Loss of Details in Simple Solutions:

- Sometimes they make the image smaller to fit the memory, but in doing so you lose small yet significant features.
- For instance, when a picture is resized it is often common to find that small objects in the background of the picture are no longer visible.

3) Multiple Classification:

- For e.g. identifying a picture as containing a dog or a cat requires understanding of whole image.
- Object detection (finding objects in an image) and segmentation (dividing an image into meaningful regions) need both fine details and a global view.

So What's The Solution ?

Here's an approach on how to tackle the mentioned challenges with feasible solutions in terms of partitioning the problem into sub-problems without a loss in performance. Here's how it works:

1) Patch-Based Processing:

- The large image is broken-up into smaller segments known as “patches”.
- To extract the details every patch is processed separately.
- This method reduces the memory needed since we process smaller parts instead of the whole image at once.

2) Adding Global Context:

- In addition to the patch processing, the system utilizes a small version of the full image as a reference.
- This “global patch” serves to orient the system to the big picture in a way that important details are not left out.

3) Smart Memory Use:

- Instead of updating all parts of the image in every step, only a few patches are updated at a time.

4) Unified Approach:

- The framework works for classification, object detection, and segmentation, which require different levels of detail.
- It uses the same core structure for all these tasks, making it versatile.

Reference(Resource Efficient Perception for Vision Systems From: A Venkata Subramanyam) - 12 May 2024

Applications:

Now that we know the importance and how image processing and recognition work, let's dive into some of its applications that we commonly use in our daily life.

1) Health Care:

Detecting brain tumors or strokes and helping people with poor eyesight are some examples of the use of image recognition in the healthcare sector. The study shows that the image recognition algorithm detects lung cancer with an accuracy of 97%.

With the increase in the ability to recognize computer vision, surgeons can use augmented reality in real operations. It can issue warnings, recommendations, and updates depending on what the algorithm sees in the operating system.

2) Banking:

Banks are increasingly using facial recognition to confirm the identity of the customer, who uses Internet banking. Banks also use facial recognition "limited access control" to control the entry and access of certain people to certain areas of the facility.

3) Manufacturing:

For pharmaceutical companies, it is important to count the number of tablets or capsules before placing them in containers. To solve this problem, Pharma packaging systems, based in England, has developed a solution that can be used on existing production lines and even operate as a stand-alone unit. A principal feature of this solution is the use of computer vision to check for broken or partly formed tablets.

Another example is a company called Sheltoncompany Shelton which has a surface inspection system called WebsPECTOR, which recognizes defects and stores images and related metadata. When products reach the production line, defects are classified according to their type and assigned the appropriate class.

4) Gaming:

Recognition models and computer vision technologies have also had a great impact on the gaming industry. It is known that the Microsoft Kinect video game is listed in the Guinness Book of Records as the fastest-selling consumer electronics device. The game is based on computer vision and tracks the human body in real-time.[5]

Conclusion:

Image recognition technology through machine learning and deep learning has revolutionized that how machines perceive and then interact with reality. From our journey to Origin of AI to complex neural networks like CNN, this field has unlocked immense possibilities.

Today Image recognition does just assist us in societal domains like Healthcare, Security and Manufacturing but also enhances our daily life in entertainment sector such as gaming.

However there are many challenges such as handling of high resolution images and memory limitations but approaches like patch-based processing and global context integration are promising solutions that continue to shape the field. With the ongoing advancement and research there would be more efficient methods to tackle these issues and drive this field forward.

Abstract:

Image recognition is a field of computer vision that focus on identifying and classifying objects within digital images. This report tells us about the principles and advanced techniques used in image recognition. We use supervised model in image recognition because of the predefined labels gives us high accuracy in image recognition in tasks such as object detection, face recognition, and handwriting analysis. The report also covers about the limitations that can be faced and also its solutions. Some of the applications of Image recognition are also highlighted.

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